



JKBOSE 2026

CLASS 11TH

PERMUTATION AND COMBINATIONS

MATHS

LECTURE-05

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TOPICS TO BE COVERED

1 What is Combination?

2 Related Problems

QUESTION

In how many ways can the letters of the word **PERMUTATIONS** be arranged if the

(i) Words start with **P** and end with **S**,

PERMUTATIONS
PERMUTATIONS

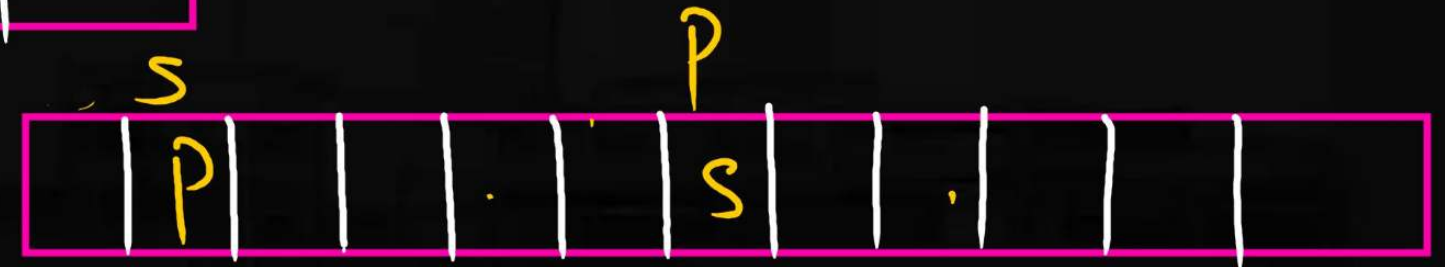
(ii) **Vowels are all together** $\left(\frac{8!}{2!} \times 5!\right)$

(iii) There are always **4 letters between P and S?**



$$\frac{10!}{2!} \times 2!$$

Jan_Meharaj1729



$$\frac{10!}{2!} \times 2!$$

Combination

Consider the letters : A, B, C.

Question: How many selections are possible by taking one at a time?

Ans) Possible selections:
A, B, C

$\Rightarrow$ No. of selections = 3

Question: How many selections are possible by taking two at a time?

$$\begin{aligned} * 3C_1 &= 3 \\ \Rightarrow 3P_1 &= 3C_1 \times 1! \end{aligned}$$

Ans) Possible selections :

AB, AC, BC

$\Rightarrow$ No. of selections = 3

$$\begin{aligned} * 3C_2 &= 3 \\ \Rightarrow 3P_2 &= 3 \times 2! \\ \Rightarrow 3P_2 &= 3C_2 \times 2! \end{aligned}$$

Question: How many selections are possible by taking all at a time?

Ans) Possible selections:
ABC

$\Rightarrow$ No. of selections = 1

$$\begin{aligned} * 3C_3 &= 1 \\ \Rightarrow 3P_3 &= 1 \times 3! \\ \Rightarrow 3P_3 &= 3C_3 \times 3! \end{aligned}$$

Definition: The selections that can be made out of a given number of things by taking some or all at a time, is known as a combination.

* $n \longrightarrow$ No. of objects (Distinct)

* ${}^n C_r$ or $\binom{n}{r}$ or $C(n, r) \longrightarrow$ No. of combinations of n objects by taking r at a time

Relation b/w ${}^n P_r$ and ${}^n C_r$:

Since, we know that each combination of ${}^n C_r$ can be permuted (arranged) in $r!$ ways.

Therefore ${}^n P_r = {}^n C_r \times r!$

$$\Rightarrow {}^n C_r = \frac{{}^n P_r}{r!} = \frac{n!}{(n-r)! r!}$$

$$\Rightarrow {}^n C_r = \frac{n!}{r!(n-r)!}, \text{ where } 0 < r \leq n.$$

* we define ${}^n C_0 = 1$

EXAMPLE



A committee of 3 persons is to be constituted from a group of 2 men and 3 women. In how many ways can this be done? How many of these committees would consist of 1 man and 2 women?

$$\begin{aligned} \text{(Sol) Total no. of ways of forming a committee} &= {}^5C_3 = \frac{5!}{3!(5-3)!} \\ &= \frac{5 \times \cancel{4} \times \cancel{3}!}{\cancel{3!} \times 2!} = 10 \end{aligned}$$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Also, the no. of ways of forming a committee

Consisting of 1 man & 2 women

$$= {}^2C_1 \times {}^3C_2 = \frac{2!}{1!(2-1)!} \times \frac{3!}{2!(3-2)!} = 2 \times \frac{3 \times \cancel{2}}{\cancel{2}} = 6$$

Hence the result.

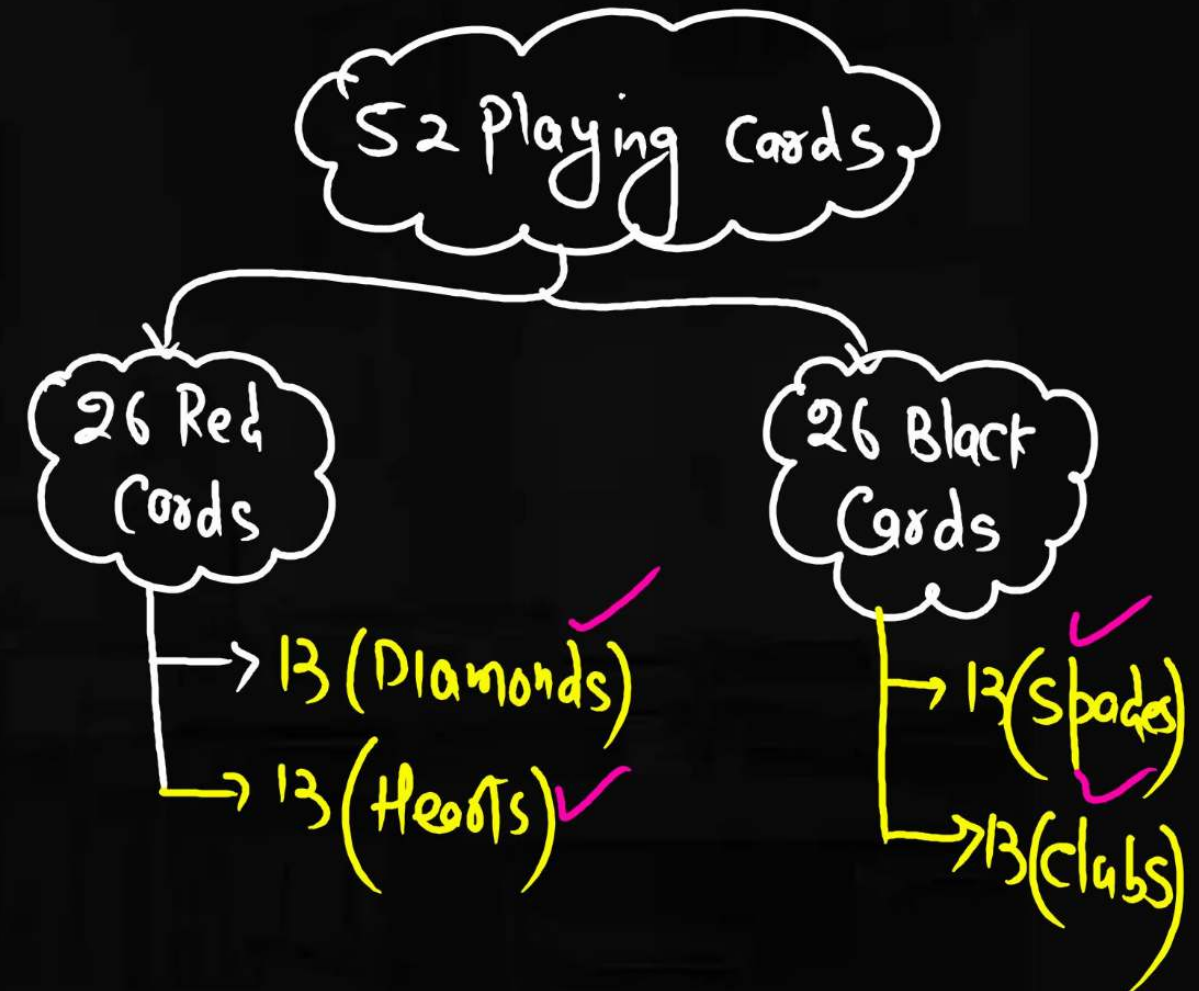
EXAMPLE

What is the number of ways of choosing 4 cards from a pack of 52 playing cards? In how many of these

- (i) Four cards are of the same suit,
- (ii) Four cards belong to four different suits,
- (iii) Are face cards,
- (iv) Two are red cards and two are black cards,
- (v) Cards are of the same colour?

* King, Queen, Jack $\rightarrow$ face Cards (12)

* King, Queen, Jack, Ace $\rightarrow$ Honour Cards (16)





$${}^nC_r = \frac{n!}{r!(n-r)!}$$

(Sol) (i) Required No. of selections
 $= {}^4C_1 \times {}^{13}C_4$

(ii) Required No. of selections
 $= {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1$

(iii) Required no. of selections $= {}^{12}C_4$

(iv) Required No. of selections
 $= {}^{26}C_2 \times {}^{26}C_2$

(v) Required No. of selections
 $= {}^2C_1 \times {}^{26}C_4$

Hence the result.

QUESTION (NCERT EXAMPLAR)

In a small village, there are 87 families, of which 52 families have atmost 2 children. In a rural development programme 20 families are to be chosen for assistance, of which atleast 18 families must have at most 2 children. In how many ways can the choice be made?

Homework #

QUESTION

If ${}^nC_2 = 45$, then n is equal to

1 10 ✓

2 45 ✗

3 11 ✗

4 9 ✗

(Sol) Given that ${}^nC_2 = 45$

$$\Rightarrow \frac{n!}{2!(n-2)!} = 45$$

$$\Rightarrow \frac{n(n-1)(\cancel{n-2})!}{2(\cancel{n-2})!} = 45$$

$$\Rightarrow \frac{n(n-1)}{2} = 45$$

$$\Rightarrow n^2 - n = 90$$

$$\Rightarrow n^2 - n - 90 = 0$$

$$\Rightarrow n^2 - 10n + 9n - 90 = 0$$

$$\Rightarrow n(n-10) + 9(n-10) = 0$$

$$\Rightarrow (n+9)(n-10) = 0$$

$$\Rightarrow n = -9, 10$$

QUESTION (NCERT EXAMPLAR)

If ${}^nC_{12} = {}^nC_8$ then n is equal to

$${}^nC_a = {}^nC_b \quad \begin{matrix} a \neq b \\ a+b=n \end{matrix}$$

1 ✓ 20

2 ✗ 12

3 ✗ 6

4 ✗ 30

(Sol) Given that ${}^nC_{12} = {}^nC_8$

$$\Rightarrow \frac{n!}{12!(n-12)!} = \frac{n!}{8!(n-8)!}$$

$$\Rightarrow 12!(n-12)! = 8!(n-8)!$$

$$\Rightarrow 12 \times 11 \times 10 \times 9 \times 8! (n-12)! = 8! (n-8)(n-9)(n-10)(n-11)(n-12)!$$

$$\Rightarrow 12 \times 11 \times 10 \times 9 = (n-8)(n-9)(n-10)(n-11)$$

$$\Rightarrow n-8 = 12$$

$$\Rightarrow n = 20$$

QUESTION

If ${}^{18}C_r = {}^{18}C_{r+2}$, then r is equal to

1

9 $\times$

2

16 $\times$

3

10 $\times$

4

8 $\checkmark$

(Sol) Given that ${}^{18}C_r = {}^{18}C_{r+2}$

$$\Rightarrow \frac{18!}{r!(18-r)!} = \frac{18!}{(r+2)!(18-r-2)!}$$

$$\Rightarrow r!(18-r)! = (r+2)!(16-r)!$$

$$\Rightarrow \cancel{r!} (18-r)(17-r) \cancel{(16-r)!} = (r+2)(r+1) \cancel{r!} \cancel{(16-r)!}$$

$$\Rightarrow 306 - 35r + \cancel{r^2} = \cancel{r^2} + 3r + 2$$

$$\Rightarrow -35r - 3r = 2 - 306$$

$$\Rightarrow -38r = -304$$

$$\frac{7}{19 \times 8} \\ \underline{152}$$

$$\Rightarrow r = \frac{304}{38} \\ \Rightarrow r = 8$$

QUESTION

If ${}^nC_1 + {}^nC_2 = 21$, then the value of n is equal to

- 1 7 ✗
 - 2 5 ✗
 - 3 21 ✗
 - 4 6 ✓
- (Sol) Given that ${}^nC_1 + {}^nC_2 = 21$
 $\Rightarrow n + \frac{n!}{2!(n-2)!} = 21$
 $\Rightarrow n + \frac{n(n-1)\cancel{(n-2)!}}{2\cancel{(n-2)!}} = 21$
 $\Rightarrow n + \frac{n(n-1)}{2} = 21$
 $\Rightarrow 2n + n^2 - n = 42$
 $\Rightarrow n^2 + n - 42 = 0$
- $\Rightarrow n^2 + 7n - 6n - 42 = 0$
 $\Rightarrow n(n+7) - 6(n+7) = 0$
 $\Rightarrow (n-6)(n+7) = 0$
 $\Rightarrow n = 6, -7$ ✗

QUESTION

If ${}^{2n}C_3 : {}^nC_2 = 44 : 3$, then the value of n is equal to

1

6 ✓ (Sol) Given that ${}^{2n}C_3 : {}^nC_2 = 44 : 3$
 $\Rightarrow \frac{(2n)!}{3!(2n-3)!} : \frac{n!}{2!(n-2)!} = 44 : 3$

2

7 ✗ $\Rightarrow \frac{(2n)!}{3!(2n-3)!} \times \frac{2!(n-2)!}{n!} = \frac{44}{3}$

3

11 ✗ $\Rightarrow \frac{2n(2n-1)(2n-2)(2n-3)!}{3 \times 2!(2n-3)!} \times \frac{2!(n-2)!}{n(n-1)(n-2)!} = \frac{44}{3}$

4

5 ✗

$$\Rightarrow \frac{2(2n-1)(2n-2)}{3(n-1)} = \frac{44}{3}$$

$$\Rightarrow \frac{2(2n-1)(n-1)}{(n-1)} = 44$$

$$\Rightarrow 2n-1 = 11$$

$$\Rightarrow 2n = 12$$

$$\Rightarrow n = 6$$

QUESTION (NCERT EXAMPLAR)

Every body in a room shakes hands with everybody else. The total number of hand shakes is 66. The total number of persons in the room is.

1 11 ~~X~~ Sol) let the no. of person = n

2 12 ✓ So, $nC_2 = 66$
 $\Rightarrow \frac{n(n-1)}{2} = 66$

3 13 ~~X~~ $\Rightarrow n^2 - n - 132 = 0$
 $\Rightarrow (n-12)(n+11) = 0$

4 14 ~~X~~ $\Rightarrow n = 12.$



QUESTION (NCERT EXAMPLAR)

The number of triangles that are formed by choosing the vertices from a set of 12 points, seven of which lie on the same line is

1

105 ✗

2

15 ✗

3

175 ✗

4

185 ✓

(Sol) Required No. of Triangles

$$= {}^{12}C_3 - {}^7C_3$$

$$= \frac{12!}{3!9!} - \frac{7!}{3!4!}$$

$$= \frac{12 \times 11 \times 10 \times 9!}{6 \times 9!} - \frac{7 \times 6 \times 5 \times 4!}{6 \times 4!}$$

$$= 220 - 35 = 185$$



QUESTION (NCERT EXAMPLAR)

The number of words which can be formed out of the letters of the word **ARTICLE**, so that vowels occupy the even place is

1 1440

2 144

3 7!

4 ${}^4C_4 \times {}^3C_3$

Homework
#

QUESTION (NCERT EXAMPLAR)

If $n_{P_r} = n_{P_{r+1}}$ and $n_{C_r} = n_{C_{r-1}}$, then the value of r is equal to:

~~1~~

~~4X~~

~~2~~

~~1X~~

~~3~~

~~3X~~

4

2 ✓

(Sol) Given that $n_{P_r} = n_{P_{r+1}}$
 $\Rightarrow \frac{n!}{(n-r)!} = \frac{n!}{(n-r-1)!}$
 $\Rightarrow (n-r)(n-r-1)! = (n-r-1)!$
 $\Rightarrow n-r = 1$
 $\Rightarrow n = r+1$ — (1)

Also $n_{C_r} = n_{C_{r-1}}$
 $\Rightarrow \frac{n!}{r!(n-r)!} = \frac{n!}{(r-1)!(n-r+1)!}$

$\Rightarrow r!(n-r)! = (r-1)!(n-r+1)!$
 $\Rightarrow r(\cancel{r-1})!(\cancel{n-r})! = (r-1)!(n-r+1)(\cancel{n-r})!$
 $\Rightarrow r = n-r+1$
 $\Rightarrow r = \cancel{r}+1-\cancel{r}+1$ (By using (1))
 $\Rightarrow r = 2$

Thank
You